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REVIEW ARTICLE

In search of coastal landscape governance: a review of its conceptualisation, operationalisation and research needs

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Abstract

Coastal landscapes are spaces and places of conflict, being among the most dynamic and vulnerable environments on the planet. Coastal landscapes are regulated through various policy instruments focused on competing interests and land uses at several intersecting administrative scales. High levels of landscape transformation, partially induced by climate change, can also be observed in several coastal countries. This article presents a systematic literature review, using the PRISMA 2020 protocol, focusing on two central and complementary debates: one focuses exclusively on coastal landscape governance (landscape research) and the other on coastal governance (coastal planning research). The systematic conceptualisation of landscape governance has emerged in the last decade, particularly in western countries. Yet, the debate to date mainly focuses on forest landscape restoration in the Global South. There has not been a comprehensive debate focused on the procedural and substantive aspects of coastal landscapes governance. Coastal governance is an emergent challenge that needs to evolve to overcome an often overstretched and fragmented governance system which has to take into account current societal challenges. Building on 174 peer-reviewed English-language articles, results reveal research trends in the conceptualisation and operationalisation of both types of governance. The review shows that there is a growing awareness of the relevance of coastal governance and landscape governance. However, evidence confirms that a robust conceptualisation and operationalisation of coastal landscape governance is missing. Conclusions point to new directions for coastal landscape governance, highlighting the need to strengthen theoretical and empirical research.

Keywords Socio-ecological systems · Landscape approach · Integrated Coastal Zone Management · Landscape sustainability · Evolutionary Governance Theory · Adaptive governance

Introduction

Coastal landscapes are spaces and places of conflict, being among the most dynamic and vulnerable environments on the planet. Coastal landscapes are regulated through various policy instruments that focus on competitive land uses and interests at several intersecting administrative scales (Rangel-Buitrago et al. 2018). High levels of coastal landscape transformation can also be observed (Romano and Zullo 2014), mainly because protecting landscapes of exceptional beauty has only been required in few coastal countries. A recent survey of the state of coastal regions worldwide demonstrated that anthropogenic pressures are severe and extensive and that "intact coastal regions are now rare" (Williams et al. 2022, p. 6). These findings corroborate a profound loss of biodiversity and landscape services and values with unprecedented environmental, economic, and social challenges ahead.

The overlapping of the different driving forces affecting coastal landscapes, and sectoral policies, instruments, and scales have substantially increased the complexity of governance (Kristensen and Primdahl 2020). Furthermore, governance challenges are expected to increase considerably in coastal countries with high infrastructure and population concentrations due to human-induced climate

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change. Indeed, the Intergovernmental Panel on Climate Change (IPCC 2019) estimated that currently, 680 million people live in a low-lying coastal zone and project that this number will increase to one billion by 2050. Even their global conservative sea-level rise projections (IPCC 2019) are highly alarming and strongly dependent on the rates of the Greenland and Antarctic ice sheets melting (Thomas and Lin 2020). Sea-level rise, extreme weather events, inundation, and erosion are some of the threats that coastal landscapes are already facing and that will intensify soon.

Governance systems must evolve to address the interactions between the system-to-be-governed and the wicked problems ahead, demanding a "more comprehensive political, social, and institutional reform than simple a symptom treatment" (Jentoft and Chuenpagdee 2009, p. 555). Governance systems must acknowledge which coastal landscapes must undergo a planned transformation, a coastal rewilding, or be involuntarily transformed by erosion and inundation processes.

Even though coastal research has become more holistic and interdisciplinary during the last four decades (Birch and Reyes 2018), landscape research has not contributed significantly to the debate. On the one hand, the concept of governance-penetrated policy and research agendas in coastal planning and management was introduced a long time ago, after the United States Coastal Zone Management Act of 1972 (Fischer 1990; Scott 1978). As a result, a wealth of literature from around the globe has been developed on coastal governance processes, structures, and practices (Partelow et al. 2020), yet the attention given to coastal landscapes remains low. In addition, a fragmented governance structure with high regulatory complexity and a lack of policy integration with inconsistent sea–land interactions to deal with coastal wicked problems remain key challenges to coastal governance in many coastal countries. On the other hand, a growing awareness of the need for more integrated public policies and planning instruments has resulted in calls for landscape-focused forms of governance, mainly because global challenges do not comply with either fixed administrative boundaries or traditional government-led jurisdictions (Arts et al. 2017; Van Oosten et al. 2021). Contemporary researchers focus on landscape as a living place, functioning as a dynamic socio-ecological system, depending on how people perceive it and recognising the need for different stakeholders involvement (Swaffield 2002; Thompson 2008). The systematic conceptualisation of landscape governance along these lines has only emerged in the last decade after the work of Görg (2007). Since Görg (2007), many scholars have called for renewed efforts towards landscape governance as a new governance prism capable of connecting contemporary global challenges with a place-based, multi-level and multi-stakeholder integrated approach.

This dichotomy between coastal and landscape governance research (coastal planning research versus landscape research) is the starting point for this systematic literature review. Our main goal is to identify potential theoretical and empirical alliances by incorporating the landscape governance lens into coastal governance to advance the coastal landscape governance emerging field. To achieve this, our review will: (1) analyse how the coastal and landscape governance debate has evolved, leading us to (2) assess the main trends and areas of discussion in both debates and (3) identify research gaps to be explored further.

The article is organised into six sections: the first section is this introduction which aims to present the need for and the main goals of this systematic review. The second section gives an overview of this review's search protocol and its limitations. In "Results" the results of coastal governance and landscape governance conceptualisation and operationalisation are presented. "Discussion on moving the emerging field of coastal landscape governance forward" presents the review discussion exploring the meaning of the results and how they could move the field of coastal landscape governance forward and, finally, "Conclusions and future research" presents the main conclusions drawn from the review.

A critical aspect is that governance, landscape, and the coastal zone assume different meanings in the literature. In order to avoid misunderstandings, in this systematic review 'governance' refers to "all processes of governing, whether undertaken by a government, market, or network, (…) whether formal or informal organization, or territory, and whether through laws, norms, power, or language" (Bevir 2012, p. 1). 'Landscape' assumes the formal definition presented by the Council of Europe (2000) on the European Landscape Convention (ELC), which claims that "landscape is defined as an area, as perceived by people, whose character is the result of the action and interaction of natural and/or human factors" (article 1). The 'coastal zone' refers to "the band of dry land and adjacent ocean space (water and submerged land) in which terrestrial processes and land uses directly affect oceanic processes and uses, and vice-versa" and it is defined at a policy level (Kay and Alder (2005, p. 2) quoting Ketchum (1972)).

Methods

Search protocol and sample

The literature review was organised and conducted using the PRISMA 2020 Protocol to meet the objectives. The search protocol was conducted in May 2021 and repeated in October 2021 using SCOPUS. The literature search started using the keywords "coastal landscape governance", "coastal governance", and "landscape governance", but no entries were

Table 1 Number of articles for each search string

Main topic	Search	Search string	Articles
Coastal landscape governance	#1	"coast* landscape" AND governance OR "coast* landscape governance"	16
	#2	"landscape governance" AND coast*	1
	#3	"landscape governance"	100
Coastal governance	#4	"coast* governance"	125
Total			242

found regarding "coastal landscape governance". Therefore, a decision was made to broaden the search to include two strings focused on coastal landscapes (search string 1: "coast* landscape" AND governance OR "coast* landscape governance") and (search string 2: "landscape governance" AND coast*). The search was limited to peer-reviewed journal papers and review articles written in English only. The period of the search was "all years" to 2021. In total, 242 records were stored in EndNote X9 software (Table 1).

All 242 articles were reviewed to confirm that the search criteria were followed (i.e., peer-reviewed scientific articles and English-language). Duplicates and no-access situations were removed, resulting in 225 records.

For each manuscript, preliminary relevance was determined by the title and abstract. If the content seemed to discuss governance processes and/or provide case studies, its complete reference was obtained. The title or abstract should also embrace the concept of holistic landscape when related to landscape governance. Otherwise, such as if the article argues that landscape was exclusively a land use or has only an ecological perspective, the hit was excluded. Furthermore, articles focused on ocean activities (e.g., fisheries management, overseas mining, coral reefs preservation) or maritime governance were excluded. Even though maritime governance has been identified as one of the critical challenges for the twenty-first century (Van Assche et al. 2020), this research is specifically interested in addressing the governance challenges to the interface of land–sea, excluding ocean governance. As a result, the review did not include the maritime spatial planning approach to the governance of seas and oceans, even though the authors recognise its relevance to both processes' interfaces.

In case of doubt, the article was always included. This process resulted in a database with 192 records. Then, the full text of each of the 192 potentially relevant hits was analysed. Here inclusion/exclusion criteria were also used. For an article to be considered relevant, it had to: (1) discuss the concept of coastal or landscape governance; if it just mentioned the concept, the article was excluded; or (2) provide case studies around the globe, where the concept has been studied or applied. A total of 174 articles (see appendix 1) fulfilled one of the criteria mentioned above (71.9% of screened articles) and were eligible for further examination. Figure 1 presents the systematic review flow diagram that synthesises the literature search methodology.

Limitations of the literature search

Several decisions were made when conducting the review which entailed associated limitations. First and conceptually, it would have been interesting to consider the evolution of coastal and landscape planning and management to track the changes these theories, concepts, and methods have undergone concerning governance and discuss their main differences and similarities. However, this would need to be a separate assessment. Nevertheless, the 174 articles reviewed were written incorporating and discussing prior research, so indirectly, these complementary developments in the literature are to some extent taken into account.

Secondly, the deliberate exclusion of the search string 'maritime governance' might have led to a different angle on coastal governance being seen. Still, the authors' research only addresses the governance of land–sea interactions.

Thirdly, the search protocol also excludes the search string 'environmental governance' because its wide scope would broaden the results and the analysis too much, yet all articles directly related to landscape or coastal systems were included.

Fourth, grey literature was excluded from the search material because this research aimed to focus exclusively on academic literature. The exclusion of plans, reports and books could lead to some bias, especially in the coastal and landscape governance operationalisation sub-sections. However, it is considered that the 174 reviewed articles are sufficient to produce a comprehensive understanding of the academic debate on coastal landscape governance.

Finally, the search only included articles written in English, retrievable via Scopus. Nevertheless, most articles (98%) cover many international authors with different scientific backgrounds.

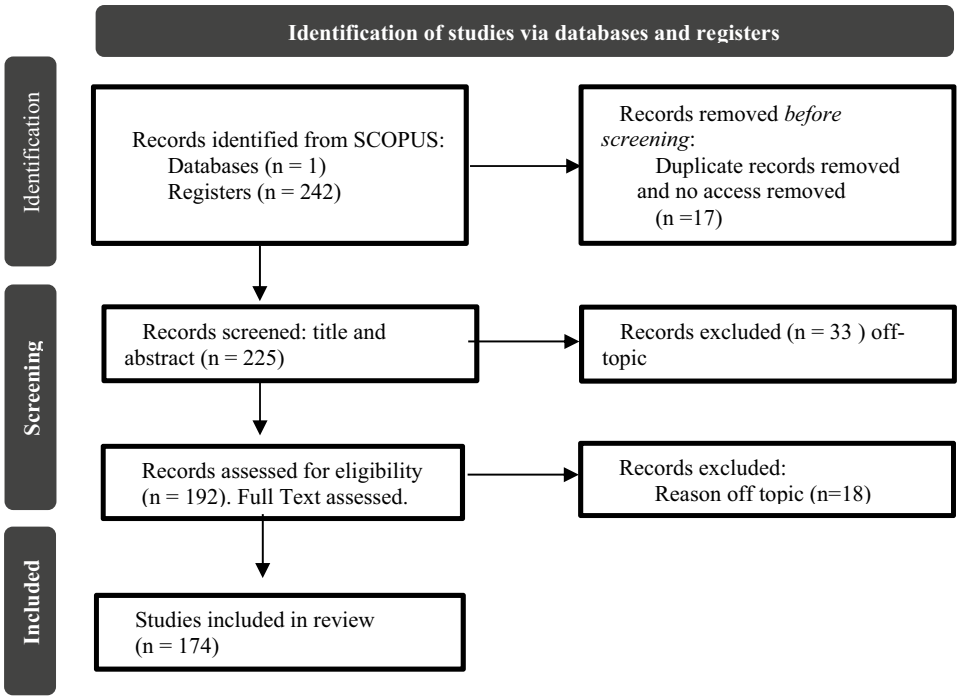

Fig. 1 Synthesis of the literature search methodology based on the PRISMA 2020 flow diagram for new systematic reviews

Results

Results on coastal governance ("Results on coastal governance") and landscape governance ("Results on landscape governance") are organised around their conceptualisation and operationalisation. 'Conceptualisation' relates to the abstraction of the concept (coastal or landscape governance) and how it has been conceived, developed, and clarified over time in theory. 'Operationalisation' refers to the procedures developed to represent and apply the concepts in the real world.

Results on coastal governance

The conceptualisation of coastal governance

The earliest publication meeting the search's selection criteria appeared in 1978, 6 years after the Coastal Zone Management Act of 1972 (US Public Law 92-583). Then, only in 2007 is there an evident increase in the number of publications, demonstrating the subject's topicality. The majority of articles (86.2%), from Europe, Oceania, North America, Africa, Asia and South America, explored a concrete governance problem.

In the literature, there seems to be no general definition of coastal governance (Table 2). Pittman and Armitage's (2016) systematic review showed that most researchers do not define it despite using the concept. Nevertheless, the authors identified three implicit trends in the conceptualisation of governance: (1) governance as a context, (2) governance as a praxis and (3) governance as a theory. Governance as a context focuses on structural components (e.g., laws, policies, procedures) and implies that management activities are a part of and strongly influenced by governance [see for instance, Flannery et al. (2015), Fletcher et al. (2014)]. Governance as a praxis focuses on the process itself (actors and powers), despite considering the structural components to also be relevant, and its relationship with management is quite unclear [see for instance, Schmidt et al. (2014)]. Governance as a theory assumes a set of ideas and hypotheses to be tested, investigated, and revised, and strongly relates to environmental governance (see Partelow et al. 2020; Van Assche et al. 2020).

The coastal governance theory debate mainly focuses on its relationship with environmental governance theories. Recently, Partelow et al. (2020) explored the central claims and abilities of various theories (polycentric, network governance, multi-level governance, collective action, governmentality, adaptive governance, interactive governance theory—IGT—and evolutionary governance theory—EGT), discussing all of them in the context of coastal systems. A key conclusion is that there will never be *a* theory of environmental governance in general or specific for the coast because every theory has its strengths and weaknesses. The authors assert that researchers must understand and use combinatory theoretical frameworks to address coastal governance. Adaptive governance, IGT and EGT were presented as broader combinatory theories, allowing a more comprehensive scope of the multiple factors influencing coastal governance. Partelow et al. (2020) show that adaptive governance

Table 2 Selected definitions of coastal governance found in the sample (adapted from Pittman and Armitage 2016)

Definition	Source
"… the process by which long-term social goals and the rules and procedures to achieve them are defined. This definition assumes governments, civil society, and markets as the principle source of power by which the processes of governance are expressed."	Pittman and Armitage (2016) Alves et al. (2013), quoting Olsen et al. (2011)
"…the formal and informal arrangements, institutions, and mores which determine how resources or an environment are utilised: how problems and opportunities are evaluated and analysed, what behavior is deemed acceptable or forbidden, and what rules and sanctions are applied to affect the pattern of resource and environmental use."	Pittman and Armitage (2016) Christie et al. (2009), quoting Juda (1999)
"… the interactions among structures, processes and traditions that determine how power is exercised, how decisions are taken, and how citizens or other stakeholders have their say."	Pittman and Armitage (2016) Cicin-Sain and Belfiore (2005), quoting Graham et al. (2003)
"…an interrelated and increasingly integrated system of formal and informal rules, rule-making systems, and actor-networks at all levels of human society (from local to global) that are set up to steer societies towards preventing, mitigating and adapting to global and local environmental change and, in particular, earth system transformation, within the normative context of sustainable development."	Pittman and Armitage (2016) Falaleeva et al. (2011), quoting Biermann et al. (2009)
"the interactions among structures, processes and traditions that determine how power and responsibilities are exercised, how decisions are taken, and how citizens and other stakeholders have their say."	Jozaei and Mitchell (2018), quoting Graham et al. (2003)
"intentional shaping of the flow of events so as to realise desired public good."	Dale et al. (2016), quoting Parker et al. (2003)

has the advantage of focusing on the relationship between the context, change and resilience, while IGT examines the interdependencies between the territory and the governance system, and EGT gives a co-evolutionary perspective on how different processes have been affecting governance.

On the governance of land–sea interaction, the uncertainty and complexity of climate change effects and the substantial changes in land use in coastal countries leads to most articles tending to reframe coastal governance through the lens of adaptive governance (Schmidt et al. 2014). Adaptive governance aims to reduce uncertainty through iterative and continual learning, and it is often discussed combined with resilience theory (Partelow et al. 2020). Lately, there has been a growing claim to combine adaptive governance with an EGT perspective (Alff 2020; Schlüter et al. 2020). EGT understands governance as a process of constant evolution (Van Assche et al. 2014) and "places governance in the context of socio-ecological systems and which understands governance and governance transformation against the background of co-evolution of all constituent parts of governance: actors, institutions, power, knowledge and narrative" (Van Assche et al. 2020, p. 1).

Van Assche et al. (2020) called for an evolutionary governance perspective to address the specificities of coastal territories, often designated as the 'coastal condition'. The coastal condition requires that the three main challenges for coastal governance must be overcome: (1) the recognition of the coast as an area itself, (2) the complexity of observation in governance between land and sea, and (3) the difficulty of policy coordination and integration on land, at sea and between land and sea. The authors assert that these three main challenges must be balanced between a global and trans-regional perspective and a local-level governance implementation (Van Assche et al. 2020), especially in decentralised coastal countries.

Despite the scientific and political debate that coastal territories must be addressed from an integrated perspective, land and sea have generally been regulated separately. Coastal territories have overlapping political/administrative boundaries with different institutions, instruments and actors and observation issues. In a modern coastal country, a specific coastal territory is commonly addressed from three political and/or administrative boundaries (national, regional and local) (Van Assche et al. 2020), increasing governance complexity to address power relations and interests. The boundaries debate crosses both governances, as a context and as praxis. Schlüter et al. (2020) highlight that the debate on governance of land–sea interactions remains superficial. Pittman and Armitage (2016) assert that there is still a need to address governance from a social–ecological approach, matching ecological and cultural contexts, boundaries, and scales. Van Assche et al. (2020) also claim that coastal governance must consider the zone rather than the boundary. The boundary must reflect the coast as a socio-ecological system which entails a local relevance, and the boundaries require multi-level governance between jurisdictions, activities, and elements. This debate gains even more relevance in a climate change scenario. Extreme weather events and

sea-level rise are not compatible with rigid boundaries of coastal zones (Pittman and Armitage 2016), reinforcing the existing problems (e.g., erosion, biodiversity loss, pollution).

Multi-level governance is also suggested concerning policy integration. Van Assche et al. (2020) indicate that a multi-governance structure is necessary. However, it must evolve to acknowledge the interdependencies at the local and regional or national levels and acknowledge the interplay between various policies, instruments, actors, and directions (land–sea). These authors also claim that new forms of policy integration cannot disregard existing forms, and their governance structure must recognise both coastal centrality and marginality. The authors highlight that policy integration may reduce the adaptive capacity claiming that "coastal governance has to observe in all directions, to grasp the interplay between actions, between policies, on land, at sea, at the coast, which all affect the coast. A high degree of coordination is thus required, a high degree of policy integration, while the identities to be coordinated (actors in and outside administrations) shall not and cannot be erased by the process of governing together" (Van Assche et al. 2020, p. 6). Coastal governance cannot depend on political cycles and must evolve to allow planning and management to adapt to change (Zafrin and Rosier 2017). Indeed, the need for empirical evidence of governance systems evolving and changing to become more effective and adaptive is recognised from different geographical contexts (Lane 2008; Schmidt et al. 2014). Another relevant theme focuses on Indigenous Coastal Management and their adaptation process to sea level rise (Abel et al. 2011). Several authors (Creed et al. 2018; Schmidt et al. 2013) concurred that adaptive coastal governance is constantly changing and evolving. However, Schmidt et al. (2013) assert that there is no particular pathway of progressive adaptation. Histories of cultural meaning and institutional bias can play an essential role in the pathways of adaptive evolution, as was proven in a Portuguese case study. Their research claimed there is a need for progression in adaptive coastal governance supported by scientific knowledge, involvement of local communities and new funding solutions.

Recently, Gonçalves et al. (2021a) also suggested that institutional interplay (i.e., the interaction between institutions) may diminish fragmentation and jurisdictional and functional gaps, enhancing cost-effectiveness. Evidence from São Paulo Macro Metropolitan Region revealed that a complete overview of institutional interplay must consider environmental conflicts, the coastal and non-coastal institutions across different jurisdictions and sectors, and the informal arrangements that shape the dynamics of the territory (Gonçalves et al. 2021a, b).

Pittman and Armitage (2016) also identified that the debate needs to involve different actors and forms of knowledge cooperating across administrative scales or jurisdictions.

Clarke et al. (2013) and Hofmeester et al. (2012) have also concurred on the benefits of including indigenous knowledge and stakeholders in coastal governance decision-making processes. The debate still focuses on specific types of knowledge (e.g., scientific knowledge) or on parts of the system to be governed (e.g., coastal ecosystems) instead of addressing a holistic perspective to improve ecological and social resources management. Knowledge production for policy-making is also named as a research need for transdisciplinary approaches (Celliers et al. 2021).

Social justice was another relevant theme, gaining more relevance due to climate change (Bennett 2019; Schmidt et al. 2013). Bennett's (2019) review asserts that coastal governance and management are highly politicised, arguing for political ecology to be engaged in coastal governance. Political ecology provides the theory and methods to engage with questions of power and politics, narratives and knowledge, scale and history, environmental justice and equity. Bennett's (2019) findings draw our attention to the need to re-politicise coastal governance and management and to the potential to expand the application of political ecology. Another relevant argument is that policy frames shape coastal governance processes, influencing how actors see the problem, the solution, and the approaches (Aukes et al. 2020). Carpenter et al. (2018) highlight that public perception research is also a relatively unexplored dimension of coastal governance, resulting in a lack of understanding of the different uses and expectations.

The operationalisation of coastal governance

Pittman and Armitage (2016, p. 9) claim that “governance is distinct from, but not exclusive of, management”. Management strategies are an essential function of governance because it represents a constant interaction between the system to be governed and governance itself.

The systematic review provided by Pittman and Armitage (2016) shows that three management approaches are relevant to addressing the coastal zone: integrated management, ecosystem-based management, and land–sea conservation planning. However, the authors reflect on an evident fragmentation between the different communities of researchers working on the different management approaches, reinforcing the need for more transdisciplinary contacts between scientific disciplines.

Integrated Coastal Zone Management (ICZM), or Integrated Coastal Management (ICM), is the most politicised approach, mainly in the Global North, but not exclusively so. It has been developed as an alternative to traditional ad-hoc and sector-based approaches that have been advocated since 1950 (Glaeser et al. 2009). After the 1970s, increased

levels of environmental awareness and the consciousness that responding to an immediate crisis was no longer the solution led to a demand for new approaches that would address the complexity of the coast. The approach explicitly tried to address coastal zones as unities, aiming for new forms of policy integration addressing both land and sea and striving for the sustainable use of coastal areas (Tabet and Fanning 2012). Glaeser et al. (2009) mentioned that ICZM is closely linked to a view of coasts as complex socio-ecological systems.

ICZM arose after the Rio de Janeiro United Nations Earth Summit (1992). The coastal states present at the event acknowledged the need to develop integrated strategies, recognising that sustainable development should assume an integrated approach for facing future challenges (Cicin-Sain and Knecht 1993). One of the most prominent characteristics of ICZM is the integration across sectors, disciplines, institutions and levels of government (Olsen 2002). As a result, ICZM has been gradually accepted as a holistic approach for coastal planning and management (Wescott 2004), with numerous countries developing ICZM policies (Caviedes et al. 2020; Lee 1998; O'Hagan and Ballinger 2009; Rosendo et al. 2018). Several initiatives followed at the European Union level to stimulate ICZM implementation in its Member States (Van Assche et al. 2020), reinforcing the EUropeanisation of national policies.

Different academic assessments followed (Fletcher et al. 2014; Khelil et al. 2019; Martino 2016; Soriani et al. 2015). However, as Glaeser et al. (2009) point out, "ICZM is characterised by a growing gap between rhetoric and practical reality at both local and global level". The concept and the success of ICZM relied on the capacity to promote integration between different policies, respond adaptively to political and societal changes, engage citizens in the decision-making process and plan against the dominant power structures (Post and Lundin 1996; Tabet and Fanning 2012). Several authors (Olsen 2002; Tabet and Fanning 2012; Van Assche et al. 2020) concluded that this approach has failed to tackle the coast as a particular zone and that actors and knowledge were also focusing on land or sea. In addition, policy integration was rarely achieved because of a lack in understanding of the coastal condition, weak governance structures and capabilities, or financial constraints. Van Assche et al. (2020) claimed that ICZM must evolve under the lens of EGT to move forward in the direction of adaptive coastal governance.

Ecosystem-based management has also gained a relevant position in the ongoing debate (Pittman and Armitage 2016), mainly linked to the ecosystem services concept (McKinley et al. 2018). Literature advocates that new management paradigms based on ecosystems must "acknowledge the multiple connections sustaining and distinguishing an ecosystem, including values and services provided by such ecosystem, the accumulated impacts and risks, the multiple and often competing management objectives, as well as [the] need to embrace change and foster social learning and adaptation from wide scale strategies to local contexts" (see the Introduction of Herbst et al. 2020). Chen et al.'s (2020) research claims that ecosystem accounting through the lens of evolutionary governance has the potential to enhance transparency, clarify material dependencies and link stocks and flows of ecosystem services and values.

Based on Álvarez-Romero et al. (2011), Pittman and Armitage (2016) stated that land–sea conservation planning correlates ecosystem-based management with integrated management. Nevertheless, its principal distinction is its focus on addressing fundamental systematic conservation planning concepts, such as complementarity, least-cost solutions, transparency, and repeatable methods.

Over time, other concepts have emerged that indicate shifts in coastal governance to address the complex interactions within coastal systems. Climate change and risk mitigation and adaptation are one theme that is gaining increasing attention in coastal governance literature (Dunning 2020; Glaser et al. 2018; Hofmeester et al. 2012; O'Riordan et al. 2014; Sauer et al. 2021). Rosendo et al. (2018) showed the need for closer integration among coastal management, disaster management and climate adaptation frameworks. O'Donnell (2019) explored the complexities of local planning policies in addressing coastal climate adaptation. Findings revealed that the most significant obstacles to climate adaptation are the possibility of having different interpretations of the same laws, inconsistencies between various legal frameworks and the prevalence of private property ideology. The urgency of governments to assume an integrated perspective, especially creating institutions that deal with the complexity of the coastal system, is also asserted (Bellanger et al. 2020; Caviedes et al. 2020). Sauer et al. (2021) demonstrated the need to articulate national policy decisions with local initiatives to foster coastal climate adaptation in the Barcelona metropolitan area to overcome how climate change adaptation is mainstreamed at different scales and sectors.

Finally, others (Jozaei and Mitchell 2018; Jozaei et al. 2020) have argued that coastal governance needs to incorporate the resilience thinking approach to address the complexity and future uncertainty of changing scenarios. The resilience thinking approach is often linked to adaptive management (Dunning 2020). Jozaei et al.'s (2020) empirical findings suggest that the resilience thinking approach is also effective in dealing with scale issues in the decision-making process and is compatible with proactive and entrepreneurial leadership.

Main findings on the coastal governance results

Since 1950, coastal planning and management have been standing as an integrative discipline. However, only in 2007 would there be a significant increase in the number of related publications. The debate focuses predominantly on Europe, Oceania and North America, and it is strongly associated with coastal management.

The discussion has mainly centred on three complementary perspectives: governance as context, as praxis, and as theory (Pittman and Armitage 2016). Coastal governance as a theory relates to abstraction of the concept and has evolved with close links to environmental governance theories. The synergies between different theories are asserted to address coastal societal challenges (Partelow et al. 2020), yet adaptive governance dominates the debate, mainly due to climate change adaptation and mitigation. Adaptive governance focuses on the context, change, learning, and monitoring process and relates to resilience theory to reduce uncertainty (Schmidt et al. 2013). Recently, EGT is also discussed as a broader theory that places governance in the context of socio-ecological systems and highlights the co-evolutionary perspective on how processes have been affecting governance, informing future governance options (Schlüter et al. 2020).

Van Assche et al. (2020) identified the three main challenges that coastal governance must address and overcome now: (1) coastal zones must be politically recognised from a socio-ecological perspective, (2) governance must address the interface between land and sea, and (3) policy integration and coordination must be accomplished.

The debate has also been addressing the structural (governance as a context) and process (governance as a praxis) components of governance, respectively, to better deal with coastal challenges. However, the debate has centred chiefly on the operationalisation of management approaches. ICZM gained a strong position after Rio 92 and aimed to address the complexity of the coastal zone from an integrated perspective. Worldwide, coastal states assumed the commitment of developing sustainable coastal policies, delivering policy integration, and addressing this dynamic ecosystem as a biophysical functional unit, tackling the interface between land and sea. At that time, ICZM governance challenges focused on approaching the coastal zone from an integrated perspective, addressing the interface between land and sea, and ensuring policy integration and coordination. Indeed, several academic assessments of ICZM policies across different geographical perspectives showed the difficulties of addressing the coast from a holistic perspective. There was a lack of understanding of the coastal condition or reduced governance structures, management capabilities or limited financial resources available (Fletcher et al. 2014; Khelil et al. 2019; Martino 2016; Soriani et al. 2015). Despite the demand for decentralised governance, there is no consensus on which institutions, instruments, and actors must be considered.

Climate change and biodiversity loss associated with global geopolitical discourses have led to significant shifts in addressing coastal governance. Ecosystem-based management linked to the ecosystem services debate has gained a relevant position in the ongoing discussions to address the complexity and uncertainty of coastal territories (McKinley et al. 2018; Pittman and Armitage 2016). On a separate note, Van Assche et al. (2020) claim that ICZM must address socio-ecological systems and EGT theories to evolve into adaptive coastal governance.

Social justice, political ecology, and public perception are also underdeveloped research themes pointed out as necessary to the strengthening of coastal governance theory and practice.

Results on landscape governance

The conceptualisation of landscape governance

In the literature, there seem to be varied definitions of landscape governance (Table 3). Kusters et al. (2020) argue that its meaning is not yet coherent even though the concept has spread since 2007 into academia, politics, and practice. Leibenath and Lintz (2018, p. 476) have also recognised that the concept is a “king of meta boundary objects" because it is constituted of two boundary concepts that have given rise to much philosophical debate between different scientific disciplines. Mansourian and Sgard (2021) discussed the diverse interpretations and applications of 'governance' concerning forest landscape restoration, highlighting that landscape governance faces higher complexity because of the multiple interpretations of the concept of 'landscape' itself when compared with environmental governance.

Indeed, researchers recognise that landscape governance evolves from environmental governance and that there is a relationship in particular with environmental policy integration (EPI) (Van Oosten et al. 2018b). However, it is argued that landscape governance complexity is higher than EPI "as it cuts across boundaries of sectors and scales", especially in countries that do not follow the socio-ecological processes of landscapes when defining their boundaries (Van Oosten et al. 2018b, p. 63).

Despite the relation between landscape and environmental governance, Leibenath and Lintz (2018) criticise that the literature, in general, is still failing in researching the relationship with other instruments and systems of governance. However, the authors recognise that Visseren-Hamakers (2015) counters this trend, analysing and discussing different concepts and approaches, looking for synergies to enhance governance to deal with complexity and fragmentation.

Table 3 Definitions of landscape governance found in the literature (Source: own compilation)

Definition	Source
"…deals with the interconnections between socially constructed spaces (the politics of scale) and 'natural' conditions of places"	Görg (2007, p. 954)
"…to refer to the actual multi-sector and multi-actor interactions at landscape level aimed at "solving societal problems and creating societal opportunities""	Ros-Tonen et al. (2014, p. 2997) based on Kooiman and Bavinck (2013)
"…concerned with the institutional arrangements, decision-making processes, policy instruments and underlying values in the system by which multiple actors pursue their interests in sustainable food production, biodiversity and ecosystem service conservation and livelihood security in multi-functional landscapes"	Kusters et al. (2020, p. 2) quoting Kozar et al. (2014)
"…the process of multi-sector, multi-actor and multi-level interaction and decision making at the landscape level."	Van Oosten et al. (2018a, b, p. 157) based on Colfer and Pfund (2011), Van Oosten et al. (2014), Ros-Tonen et al. (2015) and Kusters (2015)
"…a place-based multi-stakeholder process of negotiation and spatial decision-making within its wider institutional context, with the aim of maintaining, enhancing or restoring the landscape's functions, goods and services for the long term"	Van Oosten et al. (2021, p. 1) based on Van Oosten et al. (2014)

Although the body of literature on landscape governance is still relatively small, some common trends have arisen. Leibenath and Lintz (2018) identified two lines of conceptualising landscape governance and proposed a third one. The first followed the Görg (2007) work, the second the ideas of Kozar et al. (2014), and the proposed third line was based on Arts and Visseren-Hamdkers (2012) Triple G model (government, governance in the narrow sense and governmentality) used to analyse the forest policy field in depth. One common point in the debate is that Görg's article (2007) is identified as the starting point of systematic discussions about landscape governance, aiming to discuss the advantages of the 'landscape' concept to move forward environmental governance spatialisation in Germany. Since then, researchers worldwide have increased their interest in addressing landscape governance from a conceptual and/or empirical perspective, mainly in Europe, Africa, and Asia. Kozar et al.'s (2014) argument dominates the literature. It is centred on the governance process itself and its instruments. It has mainly been discussed in the sense of 'good governance', emphasising the "plurality and interdependency of actors, institutions and modes of landscape-related public policy-making" (Leibenath and Lintz 2018, p. 478). Many authors (Kusters et al. 2020; Van Oosten et al. 2021) have conducted empirical research to analyse the multi-level, multi-actors, and multi-sectoral approaches to collective decision-making in different geographical contexts in the Global North and South.

More recently, Leibenath and Lintz (2018) claim that, despite the fact that landscape governance has gained prominence in the literature, it still has distinct meanings. In addition, the authors criticise that both trends (Görg 2007; Kozar et al. 2014) conceptualise landscape governance as a unifying political decision-making concept, describing it as a utopian concept, where conflicts of interests are not explicitly considered as being part of landscapes. Using Germany's wind energy landscape to demonstrate it, Leibenath and Lintz (2018) challenge both Görg (2007) and Kozar et al.'s (2014) perspectives, proposing that landscape governance must be approached using three theoretical perspectives when analysing theories and empirical manifestations of landscape governance: government, governance in the narrow sense and governmentality. Although those perspectives have different epistemological assumptions, leading to different approaches to reality, their juxtaposition may strengthen the findings, leading to a more accurate perspective of power relations and effects. In 2015, Plieninger et al. (2015) also recognised the need to address environmental justice in European landscape research regarding power relations in landscape stewardship.

Power relations are also present in Görg's (2007) research, yet it is not a central theme. The author's argument comprises three critical areas as conceptualising the nature of landscape governance: (1) the relationship between the politics of scale and the natural conditions of a place, (2) the emergence of public–private actors' constellations, and (3) the policy responses. It is claimed that 'landscape' is a bridging concept that operates at different scales, representing biophysical, ecological and social scales, following the ideal stated on the ELC (Council of Europe 2000). Nevertheless, the fuzziness of the spatial scale for governance also creates different tensions and power relations (who is in or out). At the same time, it handles complex environmental and social problems fixed in a specific place with particular biophysical conditions which may yet have a global impact (Van Oosten et al. 2018a).

Furthermore, landscape governance entails a medium–large temporal scale that can represent a challenge

when dealing with socio-ecological systems. The interactions between the different components and stakeholders can operate at many scales that transcend political-administrative boundaries. These 'new spatialities' (Görg 2007) may lead to new functional spaces that may not have the legitimacy to operate. Moreover, as Van Oosten et al. (2021) demonstrated, actors in the Netherlands, Indonesia, Ethiopia, Nepal, Brazil, Rwanda and the Philippines may not have the substantive or procedural capabilities to address the '*new spatialities'* and their governance challenges.

Landscape governance boundaries assume distinct debates beyond the 'new spatialities', mainly related to property rights (Mansourian and Sgard 2021; Penker 2009). Virtual parcelling has been recently discussed by Bradshaw and Leonard (2020) in The United States of America as a possible solution to overcoming landscape governance conflicts when dealing with overlapping resources, varied ownerships and governance structures.

Although Gorg's (2007) main argument focuses on the relationship between the politics of scale and the natural conditions of a place, Leibenath and Lintz (2018) criticise the fact that the natural conditions of a place are also socially constructed. The fact that Görg (2007) emphasises the politics of scale, the neglect of the broader conceptualisation of landscape governance is also discussed. Indeed, Buizer et al. (2016) criticised the fact that the natural conditions must have an active and relational conceptualisation. Their main arguments were that the natural conditions are dynamic and that the dominant discourses and institutional practices can affect it, as was demonstrated in a Dutch agricultural landscapes case study. Nevertheless, Van Oosten et al. (2018a, p. 159), when researching large-scale commodity landscapes in Indonesia, also claimed that "the spatialisation of landscape governance requires not only new actor constellations, but also new discourses and institutional reform". In decentralized democratic countries, a 'new institutionality' is necessary to deal with landscape governance complexity, allowing public–private institutional arrangements to operate.

Another relevant distinction in the debate is introduced by Van Oosten et al. (2021). The authors discussed the differences between substantive and procedural landscape governance. The first dimension focuses on the system to be governed, aiming to maintain, enhance or restore landscape's functions, goods, and the services it provides. The second dimension looks for the governance process, seeking to enhance a legitimate place-based stakeholder process of spatial decision-making (Mansourian and Sgard 2021; Van Oosten et al. 2014, 2021). Both have a robust institutional component. Substantive landscape regulates landscape functions through different policy sectors and instruments, mainly on a sector-based perspective and at higher administrative levels of scale. Procedural landscape governance occurs across sectors, scales and jurisdictions with formal and/or informal institutional arrangements (Van Oosten et al. 2021). In 2018, Van Oosten et al.'s (2018b) research also addressed policy conflicts in Rwanda through the substantive and procedural lenses. The first is linked to conflicting policy objectives, while the second is associated with the absence of transparency and/or participatory procedures. Results showed that productive[1] bricolage, institutional bricolage[2] and institutional entrepreneurship[3] were essential for actors to address conflicting policies on landscape governance processes. Stepanova and Bruckmeier (2013) have also discussed the fact that Swedish coastal governance conflicts were mainly related to the juxtaposition of different actors representing distinct institutions with conflicting interests and perceptions.

In addition to landscape governance conflicts, other challenges have been discussed concerning landscape governance. The debate highlights the need to leave the prominent (European) 'landscape elitism', where the protection and management of the landscape are only shaped through the values of professional elites, demanding new modes of governance (Scott 2011). The fact that operationalisation must occur at the local level to be more effective is underlined because the bottom-up process for policy integration on this scale opposes established discourse, empowers institutions and produces alternatives to the business-as-usual mindset (Buizer et al. 2016). Opdam et al.'s (2016) review based on the assumption that landscape benefits impact collective action in its governance proves that more systematic and critical empirical research still needs to be conducted. Westerink et al. (2017) likewise demonstrated that introducing landscape services as a boundary concept in landscape governance in a rural case study in the Netherlands helped bridge social boundaries and enhanced social capital. Visseren-Hamakers (2015) have also emphasised the importance of debating the complexity of integrating landscape concerns into other policies compared with integration within the landscape governance system. Beunen and Opdam (2011) also discuss the need to improve the science–policy integration in European landscape research to improve local landscape governance models. Recently, Ciftcioglu's (2019) research concluded that Cyprus's coastal

[1] Productive bricolage refers to the "*flexible and dynamic crafting together of various livelihood options and its associated impacts on the landscape"* (Van Oosten et al., 2018b, p.159).

[2] Institutional bricolage refers to "*refers to the dynamic and* ad hoc *flexible nature of the governance process in the form of (re)constructing institutions, pieced together by individuals acting within the bounds of circumstantial constrains"* (Van Oosten et al., 2018b, p.159).

[3] Institutional entrepreneurship refers to a deliberate process "*to manoeuvring between sectors and scales, and to building policy networks in between*" (Van Oosten et al., 2021, p. 5).

landscape governance was unsuccessful due to a lack of integration of traditional knowledge with scientific knowledge and because planning policies still ignored landscape protection and management.

The operationalisation of landscape governance

Despite the diverse conceptualisations of landscape governance, results showed that research demands more effective and holistic approaches closely linked to landscape management (García-Martín et al. 2016; Shuttleworth 2017; Wu 2021) and landscape sustainability (Riggs et al. 2021). Scholars have advocated that landscape governance requires integrated landscape approaches to improve landscape quality and successfully support transitions towards sustainable landscape change (Albert and Vargas-Moreno 2010; Arts et al. 2017).

The debate focuses mainly on forest landscape restoration in the Global South (Bastos Lima and Persson 2020; Elbakidze et al. 2010; Riggs et al. 2018). However, other examples include natural resource management in Ghana and Burkina Faso (Foli et al. 2018), conservation and sustainable development in Ghana, Malawi, South Africa and Benin/Togo (Hedden-Dunkhorst and Schmitt 2020), rural change in Denmark (Kristensen and Primdahl 2020) and adaptation to climate change in Cameroon (Brown 2018).

Integrated landscape approaches are not 'new'. However, these approaches have become a driving paradigm in the international environmental realm during the last decade, mainly in the Global South, especially after Sayer et al. (2013). Ros-Tonen et al. (2014) recognise that landscape governance is still 'experimental', while integrated approaches have a long tradition, particularly after the 1992 Rio Summit. Integrated landscape approaches are considered the 'next generation of integrated approaches' because such approaches aim to conciliate conservation and development at the landscape scale, based on the premise that trade-offs must be identified, negotiated and accounted for, because, in any planning situation, there will be winners and losers (Reed et al. 2020).

Landscape approaches are usually created when developing countries need to conserve the most critical local resources (Arts et al. 2017). Sayer et al. (2013) proposed ten design principles for an integrated landscape approach (1) learning and adaptive management; (2) building on common concerns; (3) recognition of influences from multiple scales; (4) multifunctionality, requiring choices and trade-offs; (5) strong stakeholder engagement; (6) negotiated and transparent change logic; (7) clarification of rights and responsibilities; (8) participatory monitoring; (9) building system-wide resilience; and (10) strong capabilities of all stakeholders involved to bring together competing land uses, highlighting the need for adaptive management, stakeholder involvement and multiple objectives. The authors also identified weak governance and institutional failure as the critical implementation aspects. A landscape approach requires collaborative participation, transdisciplinary/cross-sectoral approaches, managing adaptive capacity, and applying an iterative process to address the system's inherent complexity (Arts et al. 2017). The advantages of using participatory tools to increase stakeholders' involvement have also been researched (Best et al. 2021; Ros-Tonen et al. 2021). Van Oosten et al. (2021) add that landscape approaches and landscape governance, mainly when focusing on landscape restoration, must be legitimate in the participation of the stakeholders involved (input legitimacy), the efficacy of the outcome policy (output legitimacy) and the way that the process is performed (through legitimacy).

Focusing on adaptive landscape approaches, Ros-Tonen et al. (2014) conducted a review tracking its evolution from co-management strategies and applying it to a reforestation landscape in Ghana. The authors concluded that five design principles (integrated approach, multi-stakeholder negotiation, polycentric governance, continual learning and adaptive capacity) and three enabling conditions (social capital, bridging organisations and long-term funding) must be present to achieve a truly adaptive landscape approach.

Within Europe, as one of the European research project HERCULES's conclusions, Shuttleworth (2017) claims that landscape approaches reflect the governance concept arising from the ELC. In addition, at the European level, landscape approaches represent an opportunity to apply the Convention principles on the ground and an opportunity for environmental governance implementation.

However, Ros-Tonen et al. (2018) underline that landscape approaches were not designed to contribute to the landscape governance debate. Yet, due to their synergies, they have been identified and discussed for operationalising landscape governance. A recent literature review on integrated landscape approaches in the tropics highlights its proximity to the governance process, conceptualising it as a governance strategy to achieve multi-functional and sustainable landscapes due to engagement with multiple stakeholders, and trade-off identification at the landscape scale (Reed et al. 2020).

Lastly, Ros-Tonen et al. (2018) also claim that landscape approaches are still like a jellyfish because several scholars have different interpretations despite their growing attention on the debate, reflecting the need to advance conceptual and empirical research on different geographical contexts.

Main findings on the landscape governance results

Due to the general decentralisation of government powers in democratic countries and the increased involvement of new actor constellations (Beunen and Opdam 2011), landscape

governance has been discussed in the literature as environmental governance spatialisation (Van Oosten et al. 2018b; Visseren-Hamakers 2015), which immediately evokes the concept of territorial governance.

The concept of 'landscape governance' was only introduced in 2007, after Görg (2007). Since then, the debate has been growing from a conceptual and/or empirical perspective, mainly in Europe, Africa, and Asia. Research predominantly addresses landscape restoration, especially reforestation in the Global South, yet other examples can be found worldwide.

The meaning of 'landscape governance' is still not stabilised. It is recent, and it involves two boundary concepts. Two trends dominate the debate regarding its conceptualisation (Leibenath and Lintz 2018). The first trend claims the need to reconnect governance with the politics of scale and the natural conditions of places (Görg 2007). The second dominates the literature and assesses landscape governance structures and processes (Kozar et al. 2014). Both perspectives are criticised by Leibenath and Lintz (2018) for idealising it as a utopia, where conflicts of interests and power relations are not explicitly analysed. To overcome this criticism, Leibenath and Lintz (2018) propose to address landscape governance from the perspective of government, governance in the narrow sense and governmentality. This perspective can be identified as a third way of addressing the conceptualisation of landscape governance, yet it is still in its infancy.

Recently, another relevant contribution has been made by Van Oosten et al. (2021), discussing the differences between substantive and procedural landscape governance. The substantive perspective focuses on the effects of governance on the character of the landscape, while procedural landscape governance addresses the governance process and its legitimacy.

The contemporary debate of defining landscape as a socio-ecological system has reinforced its role as a bridging concept between disciplines that occurs at multi-spatial and temporal scales, offering significant advantages for governance processes. Landscape's multi-scales for intervention, potential for integrating natural and cultural entities as well as tangible and intangible elements and capacity to offer common ground to various sectors and stakeholders in multi-level interaction are common characteristics discussed in the literature. However, a common challenge identified is landscape boundaries, as the biophysical, cultural and functional spaces may have different interpretations and may not have the necessary legitimacy to operate (Görg 2007; Van Oosten et al. 2021). Landscape conflicts (Stepanova and Bruckmeier 2013), benefits (Opdam et al. 2016) and services (Westerink et al. 2017) are also relevant themes on which the debate must evolve concerning landscape governance. The debate must also address more deeply the environmental justice research (Plieninger et al. 2015) and improve the science–practice–policy integration (Beunen and Opdam 2011).

Landscape governance can operate at multiple scales, increasing the awareness of higher or lower level processes. However, the existing institutional rigidity and the competing interests demand new approaches to deal with its complexity and move towards sustainable and multi-functional landscapes. Landscape governance is strictly related to landscape management. Integrated landscape approaches dominate landscape governance operationalisation due to their synergies (Reed et al. 2020). The debate has mainly been focusing on developing countries to conserve the most critical local resources and implement landscape restoration plans or projects (Arts et al. 2017). Several international institutions (e.g., FAO, IUCN, GLF, WWF) have also been advocating its relevance to meet societal challenges. Within Europe, integrated landscape approaches can also represent a relevant instrument to the ELC implementation (Shuttleworth 2017), yet more research is needed. Landscape approaches represent a governance strategy to address competing objectives and stakeholders, working through a transparent and legitimate process to co-produce a landscape plan, strategy, or project with greater incorporation of plural views in the decision-making process. Working at the landscape scale and identifying trade-offs and synergies contributes to discussing the tensions between development and conservation, identifying and safeguarding a landscape's unique values and identity. Integrated landscape approaches can promote democratic changes leading to more equal and fairer spaces and places.

Discussion on moving the emerging field of coastal landscape governance forward

A systematic literature review was conducted to assess the state of play and identify the research needs of two central and complementary debates between coastal governance and landscape governance. It was found that 'coastal landscape governance', the likely bridge between the aforementioned debates, has so far no entries in the SCOPUS database when applying the search protocol ("Methods"). In addition, papers on coastal and landscape governance scarcely focus on coastal landscapes. This section compares the results of coastal governance ("Results") and landscape governance ("Discussion on moving the emerging field of coastal landscape governance forward") in order to propose new directions for coastal landscape governance research.

The review demonstrated that the coastal governance debate could be traced back to 1978 (Scott 1978), but substantial growth in literature on this topic only took place after 2007. Coastal governance has predominantly discussed case studies from Europe, Oceania and North America linked to

coastal management. In contrast, we found that landscape governance was only introduced in 2007 (Görg 2007) and that the discussion has mainly focused on landscape restoration in some European, African and Asian countries. Both debates recognise the need to overcome overloaded and disintegrated governance systems (Kristensen and Primdahl 2020) to deal with high levels of unstainable landscape or land-use changes. These tendencies reflect the 'governance' debate, which has penetrated policy and research agendas worldwide and has been growing since the 1990s.

Regarding the geographical context, coastal planning and management research has been strongly influenced by the Global North's perspective, with case studies in the United Kingdom, Australia, USA, Canada and Germany leading the coastal governance debate. Although landscape research has mainly been discussed in a Global North context for a long time, the debate has expanded to the Global South to (re)frame theory and empirical research. In addition, and explicitly addressing landscape governance research, several international institutions (FAO, IUCN) have played a crucial role in politicizing the benefits governance at the landscape scale can bring to the process and to the landscape itself due to the consideration of the specific biophysical, cultural, institutional and political context. This, coupled with the fact that most African and Asian countries are leading processes to decrease their deforestation through landscape restoration and including sustainable practices on the ground, justifies most case studies focusing on this geographical context.

These results reflect the need to increase empirical research on coastal governance also from the Global South. Another evident and imperative finding is a need for theoretical and empirical research on coastal landscape governance. Coastal landscapes are one of the most dynamic and vulnerable landscapes on the planet, and research is not giving them the necessary focus to overcome the societal challenges ahead. Landscape governance must infiltrate coastal governance research.

This review showed that coastal governance and landscape governance have no consensual definition at the time of writing (Kusters et al. 2020; Pittman and Armitage 2016). As they stand today, the coast, landscape and governance are assumed as boundary concepts, the fuzziness of which contributes to their multiple meanings, with different disciplinary perspectives on their background. Despite its inherent inconsistency, the literature recognises the advantages of boundary concepts to facilitate communication and cooperation between different areas of knowledge, highlighting the need for trans-disciplinary research on landscape and coastal governance.

A common trend that our review demonstrated is that currently, the debate recognizes that the coast and the landscape are socio-ecological systems that must be governed as such and that their governance structure must be developed to address societal challenges based on this assumption (Opdam et al. 2016; Pittman and Armitage 2016; Schlüter et al. 2020; Van Assche et al. 2020).

In the landscape governance debate, the concept of *landscape* itself is also discussed as a potential advantage for governance as it calls for multi-spatial and multi-temporal scales, integrating natural, cultural, tangible and intangible elements, therefore, facilitating the effective engagement of stakeholders. The landscape concept can benefit the challenges faced in coastal governance by recognising the coast from a socio-ecological perspective and addressing the interface between land and sea (Van Assche et al. 2020). Coastal landscape characterisation and delimitation is based on its identity (Stobbelaar and Pedroli 2011), which also contributes to addressing the complexity of socio-ecological systems. The fact that landscape boundaries may also have different interpretations and may not have the political or administrative legitimacy to operate (Van Oosten et al. 2021) can be criticised. However, a landscape perspective allows greater incorporation of plural views of nature in the decision-making process, envisioning alternative futures in line with the aspirations of citizens**,** indigenous peoples, and local communities, which can legitimate the process and the outcome. Further research on coastal landscape character is necessary to identify the most relevant values that must be safeguarded in coastal landscape governance.

As our systematic review showed, coastal and landscape governance are based on environmental governance theories (Partelow et al. 2020; Visseren-Hamakers 2015). However, landscape governance is claimed as the spatial dimension of environmental governance in general and environmental policy integration in particular (Van Oosten et al. 2018a). These findings strengthen the role that landscape governance can add to coastal governance introducing a spatial dimension to it, which means that space, place and scale can be incorporated into this body of research. Coastal landscape governance must address how the coastal landscape specificities have been shaping its governance structure while assessing if its specificities have been safeguarded over time by the governance in place.

Both debates also recognise that there is a need for greater integration and synergies between different governance theories. Adaptive governance dominates the coastal governance debate, mainly related to the awareness of climate change worldwide (Schmidt et al. 2013). Learning and monitoring are essential characteristics of an adaptive process, which must be addressed in coastal landscape governance research, especially in researching for indicators to assess its adaptability.

A recent contribution was made by Van Assche et al. (2020), calling for an evolutionary governance perspective to address the coastal system. The authors place governance in the context of socio-ecological systems, claiming that an

evolutionary perspective reinforces the adaptive capabilities of the governance system. To our knowledge, landscape governance does not address evolutionary governance theory yet. However, the assessment of the co-evolution of actors, institutions, power, knowledge and narrative can also benefit the landscape governance debate. Our review also concluded that there is a need to develop empirical evidence on the evolutionary path of coastal landscape systems, analysing the performance of combined instruments on the system to be governed and on the governance process.

Another gap that our review identifies is the relationship of the debate with territorial governance. Territorial governance policies and research conceptualisation and operationalisation have influenced spatial planning throughout Europe (Van Well and Schmitt 2015). Considering the overlapping governance structures, the coastal landscape governance debate must build on these three dimensions: the landscape, the coast, and the territory. The current debate addresses policy integration of landscape and planning instruments in coastal territories (Colavitti et al. 2021; Magaudda et al. 2021), but a truly integrative perspective is still missing.

In addition, indigenous perspectives, values, knowledge, and aspirations on their coastal landscape governance and the impacts of climate change on their communities (Cisneros-Montemayor and Ota 2019) are also missing from the debate.

Findings also showed that different trends relating to coastal and landscape governance conceptualisation could be identified. The systematic review provided by Pittman and Armitage (2016) concluded that three trends could be identified in the coastal governance debate: governance as a theory, as a context, and as a praxis. Regarding landscape governance, one trend follows Görg's conceptualisation (Görg (2007), and the second the work of Kozar et al. (2014), mainly linked to the concept of 'good governance'. Leibenath and Lintz (2018) stated that power relations must be central to landscape governance and propose a new lens through which it should be seen, mainly claiming the need to combine three different perspectives: the government, the governance in the narrow sense, and the governmentality. Van Oosten et al. (2021) made a recent and relevant contribution to the debate, which advances Görg's (2007) work. According to Van Oosten et al. (2021), landscape governance is divided into substantive and procedural landscape governance. Findings demonstrate that landscape governance is mainly concerned with substantive and procedural governance components and power relations, while coastal governance research focuses on procedural (praxis and context) components. It is imperative that coastal landscape governance evolves based on both perspectives as the process and the stakeholders involved influence the landscape functions, goods and services. However, inversely, landscape character must influence its governance process too. Further research must also be conducted comparing different institutions, instruments and actors involved in coastal landscape governance.

Governance and management are not synonymous, i.e. "one function of governance is to make decisions regarding management" (Pittman and Armitage 2016, p. 10). Governance covers the collaboration of government and non-government actors, emphasising rules, processes, theories, values, institutions, actors, and discourses through which societies make decisions that affect coastal landscapes. Management covers decisions to achieve specific outcomes. Management strategies are an essential function of governance because they represent a constant interaction between the system to be governed and governance itself (Kooiman et al. 2005; Pittman and Armitage 2016). Coastal governance and coastal management are, therefore, interdependent in the sense that effective governance systems are meant to enable practical management strategies to be applied. Indeed, the coastal governance and landscape governance debates are still evolving, and regarding its operationalization, both debates have mainly focused on management approaches to operate their governance conceptualisation. Coastal and landscape governance acknowledge that governance and management must be operationalised at the local scale. In the coastal governance debate, three approaches have relevance in the literature worldwide: integrated management, ecosystem-based management and land–sea conservation planning. Integrated coastal management is the most politicised due to Rio 92, yet ecosystem-based management has become relevant to the current debate due to the growing amount of ecosystem services literature (Pittman and Armitage 2016). Integrated landscape approaches are dominant in landscape governance (Arts et al. 2017). For Reed et al. (2020) these are the 'next generation of integrated approaches' because of two intrinsic characteristics—working at the landscape scale and assuming trade-offs as a critical premise for establishing management goals. Landscape approaches have become a driving paradigm in the international environmental realm, primarily to address landscape forest restoration in the Global South.

Both ICZM and a landscape approach can be simultaneously a governance and a management process depending on the lens and scale used to assess them. ICZM's and the landscape approach's goals, values, norms and principles underpinning them are set from a governance perspective, addressing the relationship between the governance system, the system-to-be-governed, and the interactions between both. The operational tasks to achieve specific outcomes and their measurability are already in the management debate. ICZM and a landscape approach as management instruments demand a governance arrangement as a principle. However, the current debate must evolve to make a clear distinction between what constitutes coastal

landscape governance operationalisation and coastal landscape management approaches because their similarities and differences are rarely addressed in our literature sample.

Another relevant aspect on the European level is that it is also stated that landscape governance and landscape approaches contribute to implementing the ELC measures (Shuttleworth 2017), yet additional research is needed to explicitly address the correlation between them. The ELC has come into force in 40 member-states of the Council of Europe, which emphasises its central role in coastal landscape governance. The ELC's general and specific measures can reinforce coastal landscape governance implementation due to the need for landscape policies and the identification of objectives for landscape quality. Coastal landscape governance research can benefit if the debate discusses integrated coastal zone management assessments in relation to integrated landscape approaches to identify complementarities and contradictions. Their relation with indigenous-led approaches must also be explored to enrich the debate on governing and managing indigenous coastal landscapes. In addition, the way that trade-offs are assessed in landscape governance is helpful to further coastal conflicts research.

Finally, the landscape governance debate has also emphasised the need to address political ecology in empirical research, tackling questions of power and politics, narratives and knowledge, scale and history, environmental justice and equity to benefit research on coastal governance.

Conclusions and future research

This paper explored the relationships between coastal governance and landscape governance at the levels of conceptualisation and operationalisation. To achieve this goal, a systematic literature review was conducted. A considerable amount of recent literature was identified and assessed. Results have shown there is a growing awareness of the relevance of coastal governance and landscape governance. However, evidence confirms that a robust conceptualisation and operationalisation of coastal landscape governance is still missing. In the face of these knowledge gaps, many opportunities for further research in new theoretical and empirical directions were identified for coastal landscape governance.

The cross-sectoral debate between coastal governance, territorial governance and landscape governance literature can enrich how coastal systems are governed. A broad framework for analysing, comparing and promoting coastal landscape governance in policy-relevant research is fundamental for a successful adaptation to the current profound societal challenges.

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